

FIGURE NO. 04

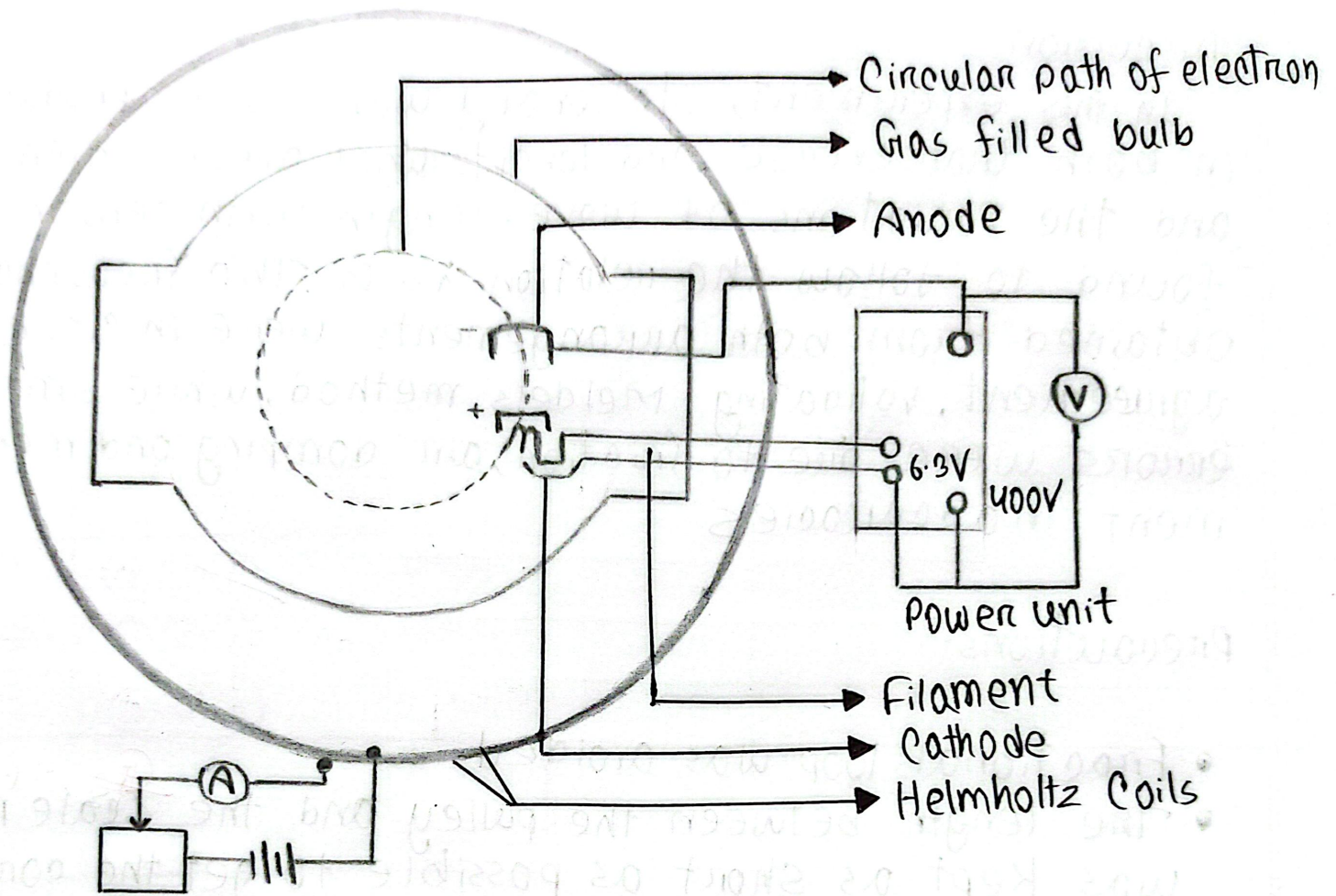


Figure 1: Electron Beam Deflection Tube

EXPT. NO. 04

DATE: 18.01.2026

EXPERIMENT OF Measurement of $\frac{e}{m}$ of an electron using an electron beam deflection tube

Theory:

When current is passed through the filament, it becomes hot and electrons gain sufficient energy to escape from the filament. This process is known as Thermionic Emission. When a charged particle moves through a region in which both electric and magnetic field are present, it experiences a force given by the Lorentz force law,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

If the charged particle is an electron of charge e moving in a magnetic field and the electric field is absent then,

$$\begin{aligned}\vec{F} &= e(\vec{v} \times \vec{B}) \\ \Rightarrow F &= e v B \sin \theta \\ \Rightarrow F &= e v B \quad [\theta = 90^\circ]\end{aligned}$$

The magnetic force provides the necessary centripetal force for circular motion, so...

$$\begin{aligned}\frac{m v^2}{r} &= e v B \\ \Rightarrow v &= \frac{e B r}{m}\end{aligned}$$

If the electron is initially accelerated by an applied potential difference V_a ,

$$\begin{aligned}e V_a &= \frac{1}{2} m v^2 \\ \Rightarrow \frac{e}{m} &= \frac{2 V_a}{B^2 r^2}\end{aligned}$$



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The magnetic field is produced by a pair of Helmholtz Coils, which generates a nearly uniform magnetic field near the center. The magnetude of the field is,

$$B = \mu_0 \left(\frac{4}{5}\right)^{3/2} \frac{nI}{a} = 8.99 \times 10^{-7} \frac{nI}{a}$$

Where,

n = number of turns in each of the pair of coils = 132

I = current in the coil

a = radius of the coil = 0.1475

μ_0 = permeability of the empty space (vacuum)
 $= 4\pi \times 10^{-7} \text{ Wb A}^{-1} \text{ m}^{-1}$

Required Apparatus:

- (i) Electron Beam deflection tube with Helmholtz Coils
- (ii) Power supply

Procedure:

- (i) The apparatus was placed on a level table, connected to AC power and warmed up for 20 minutes
- (ii) The Accelerating Voltage (V_a) was varied from 0-275 by rotating the voltage knob and a magnetic field was applied using the Helmholtz Coil to observe the circular electron beam



$r^2 \sim V_a$ graph

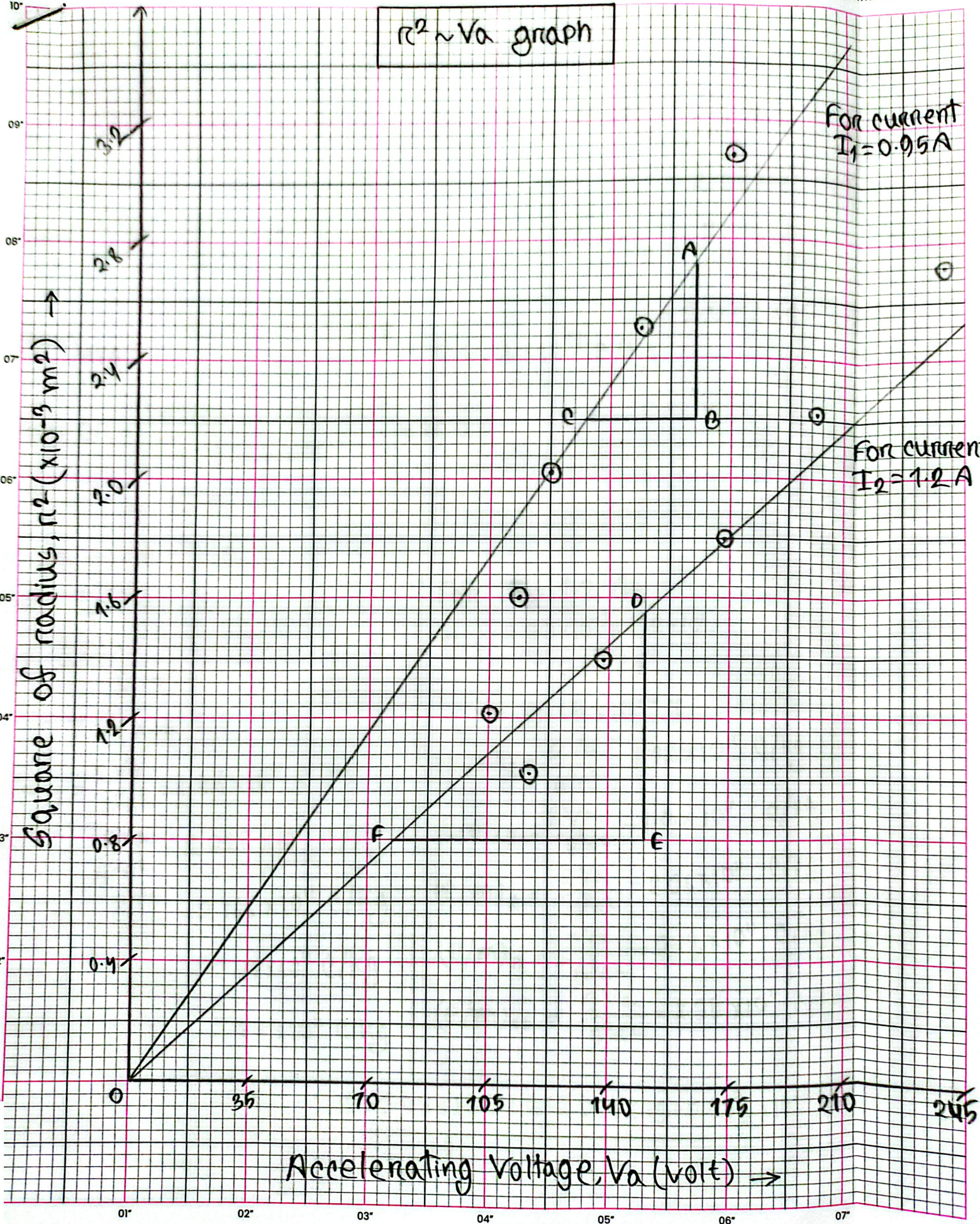


FIGURE NO.

calculations:

From graph,

$$m_1 = \frac{AB}{BC} = \frac{0.52}{31.5} \times 10^{-3} = 0.0165 \times 10^{-3} ; (e/m)_1 = \frac{2}{B^2} \times \frac{1}{m_1} = 2.08 \times 10^{11} \text{ C/kg}$$

$$m_2 = \frac{DE}{EF} = \frac{0.76}{73.5} \times 10^{-3} = 0.0103 \times 10^{-3} ; (e/m)_2 = \frac{2}{B^2} \times \frac{1}{m_2} = 2.077 \times 10^{11} \text{ C/kg}$$

$$(e/m)_{\text{avg}} = \frac{(e/m)_1 + (e/m)_2}{2} = 2.079 \times 10^{11} \text{ C/kg}$$

$$\therefore \text{Error} = \frac{|1.7588 - 2.079| \times 10^{11}}{1.7588 \times 10^{11}} \times 100 \% = 18.2 \%$$

from calculation,

For $I_f = 0.95 \text{ A}$,

$$B = \frac{8.99 \times 10^{-7} \times 132 \times 0.95}{0.1475}$$

$$= 7.64 \times 10^{-4} \text{ T}$$

$$(e/m)_1 = \frac{2V_a}{B^2 r^2} = 1.98 \times 10^{11} \text{ C/kg}$$

$$(e/m)_2 = \frac{2V_a}{B^2 r^2} = 2.028 \times 10^{11} \text{ C/kg}$$

$$(e/m)_3 = \frac{2V_a}{B^2 r^2} = 2.166 \times 10^{11} \text{ C/kg}$$

$$(e/m)_4 = \frac{2V_a}{B^2 r^2} = 2.42 \times 10^{11} \text{ C/kg}$$

$$(e/m)_5 = \frac{2V_a}{B^2 r^2} = 2.94 \times 10^{11} \text{ C/kg}$$

For $I_f = 1.2 \text{ A}$,

$$B = \frac{8.99 \times 10^{-7} \times 132 \times 1.2}{0.1475}$$

$$= 9.65 \times 10^{-4} \text{ T}$$

$$(e/m)_6 = \frac{2V_a}{B^2 r^2} = 1.88 \times 10^{11} \text{ C/kg}$$

$$(e/m)_7 = \frac{2V_a}{B^2 r^2} = 1.9 \times 10^{11} \text{ C/kg}$$

$$(e/m)_8 = \frac{2V_a}{B^2 r^2} = 2.06 \times 10^{11} \text{ C/kg}$$

$$(e/m)_9 = \frac{2V_a}{B^2 r^2} = 2.13 \times 10^{11} \text{ C/kg}$$

$$(e/m)_{10} = \frac{2V_a}{B^2 r^2} = 2.35 \times 10^{11} \text{ C/kg}$$

$$(e/m)_{\text{avg}} = \frac{(1.98 + 2.028 + 2.166 + 2.42 + 2.94 + 1.88 + 1.9 + 2.06 + 2.13 + 2.35) \times 10^{11}}{10} = 2.1854 \times 10^{11} \text{ C/kg}$$

$$\therefore \text{Error} = \frac{(1.7588 - 2.1854) \times 10^{11}}{1.7588 \times 10^{11}} \times 100 \% = 24.26 \%$$

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- (iii) For a fixed coil current ($I = 0.95 \text{ A}$), the beam diameter was measured at different accelerating voltages (V_a) five times
- (iv) The process was repeated again for another fixed coil current ($I = 1.2 \text{ A}$) by varying the V_a five times
- (v) Graphs of r^2 vs V_a were plotted and the value of e/m was determined from the slope for both coil currents

Experimental Data:

Table 1: Data for determination of e/m for different coil current

Coil current, I (A)	Magnetic field, B (T)	Accelerating Voltage, V_a (V)	Diameter, d (m)	radius, $r = d/2$ (m)	r^2 (m^2) $\times 10^{-3}$	e/m (C/Kg) $\frac{2\bar{V}_a}{B^2 r^2}$
0.95	7.64×10^{-4}	175	0.11	0.055	3.025	1.98×10^{11}
		148	0.10	0.05	2.5	2.028×10^{11}
		128	0.09	0.045	2.025	2.166×10^{11}
		113	0.08	0.04	1.6	2.42×10^{11}
		105	0.07	0.035	1.225	2.94×10^{11}
1.2	9.65×10^{-4}	241	0.105	0.0525	2.76	1.88×10^{11}
		200	0.095	0.0475	2.26	1.9×10^{11}
		173	0.085	0.0425	1.8	2.06×10^{11}
		139	0.075	0.0375	1.4	2.13×10^{11}
		116	0.065	0.0325	1.06	2.35×10^{11}

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Results:

$e/m = 2.185 \times 10^{11} \text{ C/kg}$	... (from calculation)
$e/m = 2.079 \times 10^{11} \text{ C/kg}$	... (from graph)
% of error = 24.26	... (from calculation)
% of error = 18.2	... (from graph)

Discussion:

In this experiment, the charge to mass ratio (e/m) of the electron was determined using magnetic deflection in a Helmholtz Coil arrangement. The calculated e/m value of $2.185 \times 10^{11} \text{ C/kg}$ showed 24.26% error, while the graphical method yielded $2.079 \times 10^{11} \text{ C/kg}$ with 18.2% error.

Precautions:

- Care was taken to determine B as accurately as possible
- The path of the electrons was made as circular as possible
- Parallax error was avoided
- The anode voltage was not raised higher than necessary. Usually a few hundred volts (max 275V) was enough